Firm Heterogeneity: CES, Melitz, and Hopenhayn

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Motivation: Why Firm Heterogeneity?

- Thus far: representative firms
  - Within narrowly defined industries, firms differ enormously in size, productivity, and survival
- Some canonical stylized facts:
  - Firm size distributions are highly skewed
  - Most firms are small; a small number of large firms account for disproportionate share of output and employment
  - Entry and exit are pervasive, even in "normal" times
  - Exporters are systematically larger and more productive than non-exporters

Motivation: Why It Matters for Macro

- Aggregate productivity is not just about how productive the average firm is
  - It also depends on how resources are *allocated* across firms of different productivity
- Reallocation — resources moving from low- to high-productivity firms — can be a first-order source of aggregate growth
  - Entry of productive firms, exit of unproductive ones
  - Trade liberalization reallocating market share toward the most productive exporters
- Policy distortions that are uncorrelated with productivity can misallocate resources and lower aggregate TFP, even with no change in any individual firm's technology
- Preview: this is exactly the logic behind both models we will study

CES Preferences: Motivation

- Almost every heterogeneous-firm model needs a demand system with two properties:
  - *Love of variety*: consumers value having access to more distinct goods
  - *Tractable firm-level demand*: each firm should face a simple residual demand curve, so its problem is a simple optimization
- The **constant elasticity of substitution (CES)** aggregator, due to Dixit and Stiglitz (1977), delivers both
- It underlies monopolistic competition models throughout trade, growth, and macro — worth deriving carefully once

CES Preferences: Setup

- Suppose there is a continuum (or countable set) of differentiated varieties $\omega \in \Omega$, each with consumption $q(\omega)$
- The representative consumer aggregates varieties into a single consumption index Q:

$$Q = \left(\int_{\Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} \, d\omega \right)^{\frac{\sigma}{\sigma-1}}$$

  - $\sigma > 1$ is the *elasticity of substitution* across varieties
  - Larger σ: varieties are closer substitutes
  - As $\sigma \to \infty$: varieties become perfect substitutes (linear aggregator)
  - As $\sigma \to 1$: aggregator approaches Cobb-Douglas

CES Preferences: The Consumer's Problem

- The consumer has total expenditure E to allocate across varieties, taking prices $p(\omega)$ as given
- Two equivalent ways to think about this:
  - Maximize Q subject to $\int_\Omega p(\omega)q(\omega)\,d\omega \leq E$
  - Minimize expenditure $\int_\Omega p(\omega)q(\omega)\,d\omega$ subject to achieving a target Q
- We'll use expenditure minimization, as it gives us the price index in one step. Homework: do it the other way
- Whiteboard

CES Preferences: Expenditure Minimization

$$\min_{\{q(\omega)\}} \int_\Omega p(\omega)q(\omega)\,d\omega \quad s.t. \quad \left(\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}}\,d\omega\right)^{\frac{\sigma}{\sigma-1}} \geq Q$$

- Lagrangian:

$$\mathcal{L} = \int_\Omega p(\omega)q(\omega)\,d\omega + \mu\left[Q - \left(\int_\Omega q(\omega)^{\frac{\sigma-1}{\sigma}}\,d\omega\right)^{\frac{\sigma}{\sigma-1}}\right]$$

- FOC with respect to $q(\omega)$ for a given variety ω:

$$p(\omega) = \mu \cdot Q^{\frac{1}{\sigma}} \cdot q(\omega)^{-\frac{1}{\sigma}}$$

CES Preferences: Relative Demand

- From the FOC, for any two varieties ω, ω':

$$\frac{p(\omega)}{p(\omega')} = \left(\frac{q(\omega)}{q(\omega')}\right)^{-\frac{1}{\sigma}} \implies \frac{q(\omega)}{q(\omega')} = \left(\frac{p(\omega)}{p(\omega')}\right)^{-\sigma}$$

- Relative demand depends only on relative price, with constant elasticity σ
  - This is where the "constant elasticity of substitution" name comes from
  - A 1% relative price increase lowers relative quantity demanded by σ%
- Note: changes in prices of a third good ω'' does not impact my *relative* demand for ω and ω'
- We still need to pin down the *level* of demand for each variety — this requires the price index

CES Preferences: The Price Index

- Substitute the FOC back into the definition of Q and solve for the multiplier μ
- Result: μ equals the **CES price index**:

$$P \equiv \left(\int_{\Omega} p(\omega)^{1-\sigma} \, d\omega \right)^{\frac{1}{1-\sigma}}$$

  - P is the minimum expenditure required to buy one unit of the composite good Q

- Total expenditure satisfies $E = PQ$, exactly as in a single-good problem

CES Preferences: Demand Function

- Combining the FOC, the price index, and $E = PQ$ delivers the demand for an individual variety:

$$q(\omega) = \left(\frac{p(\omega)}{P}\right)^{-\sigma} Q = \left(\frac{p(\omega)}{P}\right)^{-\sigma} \frac{E}{P}$$

- Two key properties for what follows:
  - Demand for variety ω depends on its own price *relative to the aggregate price index*, with constant elasticity σ
  - A single firm producing one variety ω treats P and E (or Q) as given
    - Idea: Firms think only about aggregate variables when determining their own optimal quantity and price

CES and Monopolistic Competition: The Firm's Problem

- Consider a firm producing a single variety ω using labour at wage w with constant marginal cost and productivity φ: $c(\varphi) = w/\varphi$
- The firm is a monopolist over its own variety, but takes P and aggregate expenditure E as given
- Facing demand $q(\omega) = (p(\omega)/P)^{-\sigma} E/P$, the firm chooses $p(\omega)$ to maximize profit:

$$\max_{p(\omega)} \left[p(\omega) - \frac{w}{\varphi} \right] q(\omega)$$

CES and Monopolistic Competition: Constant Markup Pricing

- As derived last semester, price is a constant markup over marginal cost:

$$p(\varphi) = \frac{\sigma}{\sigma - 1} \cdot \frac{w}{\varphi}$$

- Implications:
  - More productive firms (*higher* φ) charge *lower* prices
  - The markup $\sigma/(\sigma - 1)$ is the same for every firm — this is a special (and sometimes counterfactual) feature of CES. Often called "CES markup"
  - Revenue, profit, and employment are all monotonic, tractable functions of φ

CES: Revenue and Profit as Functions of Productivity

- Exercises to check for homework: revenue

$$r(\varphi) = E\, P^{\sigma-1} \left(\frac{\sigma}{\sigma - 1} \frac{w}{\varphi} \right)^{1-\sigma}$$

- Profits:

$$\pi(\varphi) = \frac{r(\varphi)}{\sigma}$$

Bridge to Heterogeneous-Firm Models

- We now have everything we need from the demand side:
  - A tractable mapping $\varphi \mapsto \pi(\varphi)$ from firm productivity to profit
  - Firms only need to know their own φ (and aggregates P, E, w) to solve their problem
- What CES does *not* give us:
  - Where φ comes from, and how it evolves over time (Hopenhayn)
  - Which firms are active in which markets (Melitz)
- We will do the above in the reverse order

Melitz (2003): Motivation

- Pre-Melitz trade models (e.g., Krugman 1980) had monopolistic competition with CES, but *identical* firms
  - Every firm in an industry either exports, or none do
- This is sharply at odds with the data: within any industry, only some firms export, and exporters are larger and more productive
- Melitz (2003) combines:
  - CES monopolistic competition
  - Heterogeneous, ex-ante-uncertain firm productivity, with entry/exit
  - A fixed cost of exporting

  to generate self-selection into exporting based on productivity

Melitz: Environment

- Monopolistically competitive industry, CES preferences with elasticity $\sigma > 1$
- A continuum of potential entrants, as in Hopenhayn: pay sunk cost c_e to draw productivity φ from distribution $g(\varphi)$
- After observing φ, a firm decides whether to:
  - Exit immediately without producing
  - Produce for the domestic market only, paying fixed cost f_d
  - Produce for the domestic market *and* export, additionally paying fixed cost $f_x > f_d$
    - Exporting incurs a marginal "iceberg" trade cost of $\tau > 1$
    - In order to deliver x units of output to the foreign market, must ship τx
- Static model for now – original paper is dynamic, but not as important

Melitz: Revenue and Profit, Revisited

- From earlier domestic operating profit as a function of productivity is:

$$\pi_d(\varphi) = \frac{r_d(\varphi)}{\sigma} - f_d, \qquad r_d(\varphi) = EP^{\sigma-1}\left(\frac{\sigma}{\sigma-1}\frac{w}{\varphi}\right)^{1-\sigma}$$

- Exporting is symmetric, but revenue is earned in the foreign market (size E^*, price index P^*) and marginal cost is inflated by the iceberg cost τ:

$$\pi_x(\varphi) = \frac{r_x(\varphi)}{\sigma} - f_x, \qquad r_x(\varphi) = E^*(P^*)^{\sigma-1}\left(\tau\frac{\sigma}{\sigma-1}\frac{w}{\varphi}\right)^{1-\sigma}$$

  - Both $r_d(\varphi)$ and $r_x(\varphi)$ are increasing in φ — higher productivity firms earn more revenue in *both* markets

Melitz: The Domestic Exit Cutoff

- A firm producing for the domestic market only operates if and only if $\pi_d(\varphi) \geq 0$
- Since $\pi_d(\varphi)$ is strictly increasing in φ, there is a unique **domestic cutoff** φ_d^* solving

$$\pi_d(\varphi_d^*) = 0$$

  - Firms with $\varphi < \varphi_d^*$ never produce — they exit immediately after drawing their productivity
  - This is the **Zero Profit Cutoff** for the domestic market

Melitz: The Export Cutoff

- Conditional on producing domestically, a firm additionally exports if and only if $\pi_x(\varphi) \geq 0$
- This defines an **export cutoff** φ_x^* solving

$$\pi_x(\varphi_x^*) = 0$$

- Generally, we want the export cutoff to be higher than the domestic production cutoff, but this does not need to be the case
  - Result: only the *most productive* firms self-select into exporting
  - This is exactly the exporter productivity premium observed in the data. Emerges endogenously, rather than being assumed

Melitz: The Three Firm Types

- In equilibrium, firms sort into three groups by productivity:
  - $\varphi < \varphi_d^*$: exit immediately, never produce
  - $\varphi_d^* \leq \varphi < \varphi_x^*$: produce for the domestic market only
  - $\varphi \geq \varphi_x^*$: produce domestically *and* export
- Total profit for an active firm:

$$\pi(\varphi) = \pi_d(\varphi) + \mathbb{1}\{\varphi \geq \varphi_x^*\}\, \pi_x(\varphi)$$

Melitz: Free Entry

- Potential entrants pay sunk cost c_e to draw $\varphi \sim g(\varphi)$
- In the (simplest) static version with a constant per-period probability of death δ, expected discounted profit from entry is equal to the cost to enter (free entry):

$$\frac{1}{\delta} \int_{\varphi_d^*}^{\infty} \pi(\varphi)\, g(\varphi)\, d\varphi = c_e$$

Melitz: Selection

- $g(\varphi)$ is the *ex-ante* distribution: the productivity draw available to any entrant before the exit decision
- The distribution of productivity *among active firms* is the truncation of g at φ_d^*:

$$\mu(\varphi) = \frac{g(\varphi)}{1 - G(\varphi_d^*)}, \qquad \varphi \geq \varphi_d^*$$

- This truncation is the model's version of *selection*: only firms that clear the cutoff are observed operating, so the observed (ex-post) productivity distribution is shifted up relative to the underlying draw distribution

Melitz: Selection cont.

- Note that changes in cutoffs $\varphi_d^\star, \varphi_x^\star$ will change the distribution of productivity (size) among surviving firms, even if the firms themselves did not change!
- Selection can lead to some interesting counterfactuals, particularly in GE

Melitz: Trade Liberalization and Selection

- Consider a fall in τ (or f_x) — trade becomes easier
  - Let's assume there is a symmetric foreign market, who also sees a reduction in trade costs
- Direct effect: φ_x^* falls — more firms find it profitable to export
- Note: average productivity of exporting firms falls!

Melitz: Trade Liberalization and Selection cont.

- General equilibrium effect: tougher foreign competition
  - Entering foreign firms lowers domestic price index P, lowering revenue conditional on φ
  - Increased demand for labour from expanding exporters bids up the wage w
  - Least productive domestic-only firms, who cannot cover the higher wage, are pushed out
- Net effect: resources reallocate *from* the least productive firms *to* the most productive (exporting) firms — this reallocation raises aggregate (industry) productivity, even holding each firm's own technology fixed

Melitz: Welfare

- Trade liberalization in this model raises welfare through channels beyond the traditional gains from trade:
  - *Selection effect*: aggregate productivity rises as resources reallocate to more productive firms
  - *Variety effect*: consumers gain access to a larger set of foreign varieties (love of variety from CES)
    - Note: also lose some domestic varieties as low-productivity domestic firms exit!
  - *Competition effect*: entering foreign firms drive down prices P

Melitz: Equilibrium Summary

- A (static) Melitz equilibrium is characterized by:

  1. Zero profit cutoff: $\pi_d(\varphi_d^*) = 0$
  2. Export cutoff: $\pi_x(\varphi_x^*) = 0$
  3. Free entry condition: $\frac{1}{\delta} \int_{\varphi_d^*}^{\infty} \pi(\varphi) g(\varphi)\, d\varphi = c_e$
  4. Labour market clearing

Melitz: Computation

- To solve Melitz on the computer, need to specify a primitive productivity distribution G
- Chaney (2008): the Pareto distribution gives closed form solutions to many key objects
  - Result was important enough to get Chaney an AER. I will assign some parts of this derivation as homework, it is somewhat painful
- From here, computation proceeds as follows:
  - Guess P, and normalize $w = 1$
  - Solve for production, export cutoffs $\varphi_d^\star, \varphi_x^\star$ given fixed costs
  - Use the free entry condition to pin down the mass of firms
  - Update the price index given the mass of firms, and wages using labour market clearing
  - Repeat until convergence

Hopenhayn (1992): Motivation

- Question: what determines the equilibrium distribution of firm size, and the rates of entry and exit in an industry?
- Hopenhayn (1992) builds the canonical model of firm dynamics:
  - Firms are ex-ante identical but draw idiosyncratic productivity shocks
  - Firms enter, operate, and exit endogenously
  - A stationary equilibrium pins down the entire firm size distribution

Hopenhayn: Environment

- A continuum of differentiated varieties ω, with CES demand and elasticity $\sigma > 1$
  - Original Hopenhayn: perfect competition with undifferentiated varieties, firms take prices as given.
  - Profits, production function left general
- Demand for an individual variety:

$$q(\omega) = \left(\frac{p(\omega)}{P}\right)^{-\sigma} \frac{E}{P} = EP^{\sigma-1}p(\omega)^{-\sigma}$$

- Each firm has idiosyncratic productivity φ, which follows a Markov process $P(\varphi' \mid \varphi)$
- Production has constant returns:

$$q = \varphi n$$

Hopenhayn: Timing

- Within a period, for an incumbent firm with current state φ:
  1. Observe current productivity φ
  2. Choose labour n, produce, earn static profit $\pi(\varphi)$
  3. Decide whether to exit (permanently) or continue
  4. If continuing, next period's productivity φ' is drawn from $P(\varphi' \mid \varphi)$
- Potential entrants: pay a sunk entry cost c_e to draw an initial productivity from some distribution $\nu(\varphi)$, then behave as incumbents from the next period on

Hopenhayn: Static Profit

- As before, firms take aggregate expenditure, the price index, and wages as given
- A firm with productivity φ faces

$$q(\varphi) = EP^{\sigma-1}p(\varphi)^{-\sigma}$$

  and has variable cost $wn = wq/\varphi$

- The firm's static problem is

$$\max_{p} \left(p - \frac{w}{\varphi}\right) EP^{\sigma-1}p^{-\sigma} - wc_f$$

Hopenhayn: CES Pricing and Revenue

- CES demand implies a constant markup over marginal cost:

$$p(\varphi) = \frac{\sigma}{\sigma - 1} \frac{w}{\varphi}$$

- Revenue:

$$r(\varphi) = p(\varphi) q(\varphi) = E P^{\sigma - 1} \left(\frac{\sigma}{\sigma - 1} w \right)^{1 - \sigma} \varphi^{\sigma - 1}$$

Hopenhayn: CES Static Profit

- Variable profit is a constant share of revenue:

$$\pi_v(\varphi) = \frac{r(\varphi)}{\sigma} = \frac{1}{\sigma} E P^{\sigma-1} \left(\frac{\sigma}{\sigma - 1} w \right)^{1-\sigma} \varphi^{\sigma-1}$$

- Including the fixed operating cost c_f in units of labour:

$$\boxed{\pi(\varphi) = \frac{E P^{\sigma-1}}{\sigma} \left(\frac{\sigma}{\sigma - 1} w \right)^{1-\sigma} \varphi^{\sigma-1} - w c_f}$$

Hopenhayn: The Incumbent's Bellman Equation

- Let $V(\varphi)$ denote the value of an incumbent firm with current productivity φ, before the exit decision
- The Bellman equation is

$$V(\varphi) = \pi(\varphi) + \max\left\{0,\ \beta \int V(\varphi')P(d\varphi' \mid \varphi)\right\}$$

- The CES demand system enters the dynamic problem only through $\pi(\varphi)$
  - This is otherwise the same recursive entry/exit problem as Hopenhayn

Hopenhayn: The Exit Decision

- Define the continuation value

$$W(\varphi) \equiv \beta \int V(\varphi')P(d\varphi' \mid \varphi)$$

- The firm exits if and only if $W(\varphi) < 0$
- If $P(\varphi' \mid \varphi)$ is "stochastically increasing" in φ, then $W(\varphi)$ is increasing in φ
  - This implies exit is characterized by a cutoff φ^*
  - φ^* solves $W(\varphi^*) = 0$
- Intuition: low-productivity firms expect future profits to be low (persistence)

An Aside: First Order Stochastic Dominance

- What does "stochastically increasing" mean? Formally, a sufficient condition for the above is *first order stochastic dominance*: if $\varphi_2 > \varphi_1$, then

$$P(\varphi' \leq z|\varphi_2) \leq P(\varphi' \leq z|\varphi_1)$$

- Starting from a higher initial productivity makes every sufficiently high productivity at least as likely

- A realistic assumption: productivity ranking of firms not likely to invert

Hopenhayn: Free Entry

- Each potential entrant pays sunk cost c_e to draw an initial productivity $\varphi_0 \sim \nu(\varphi)$, then becomes an incumbent next period

- Because entry is paid today and the incumbent value begins next period:

$$\boxed{wc_e = \beta \int V(\varphi)\nu(\varphi)d\varphi}$$

- Higher aggregate demand raises the value of entry:

$$EP^{\sigma-1} \uparrow \quad \Rightarrow \quad \pi(\varphi) \uparrow \quad \Rightarrow \quad V(\varphi) \uparrow$$

- Free entry disciplines the aggregate demand conditions and the mass of firms

Hopenhayn: Aggregate Feedback

- In competitive Hopenhayn, the industry price p clears the goods market
- With CES demand, the analogous aggregate object is $EP^{\sigma-1}$ which depends on the CES price index P
- Firm-level demand and profits therefore depend on the entire firm size distribution μ:

$$\mu \to P \to \pi(\varphi), V(\varphi) \to \varphi^*, \mu$$

- Free entry and market clearing jointly determine the aggregate equilibrium, rather than a single competitive-good price

Hopenhayn: The Stationary Distribution

- Let $\mu(\varphi)$ denote the **mass** of active firms with productivity φ, and M the mass of entrants each period
- In a stationary equilibrium, the distribution is constant over time: surviving incumbents plus new entrants replace firms that exit
- Law of motion:

$$\mu'(\varphi') = \int_{\varphi \geq \varphi^*} P(\varphi' \mid \varphi)\mu(\varphi)d\varphi + M\nu(\varphi')$$

- Distribution now affects the CES price index, so the firm distribution feeds back into demand

Hopenhayn: Labour Market Clearing

- One remaining equilibrium condition: the labour market clears
- Since μ already contains the active firms:

$$\boxed{M c_e + \int (n(\varphi) + c_f)\, \mu(\varphi) d\varphi = L}$$

- This condition pins down the scale of the active-firm distribution, given the equilibrium wage and demand conditions
- With $q(\varphi) = \varphi n(\varphi)$, firm-level employment is

$$n(\varphi) = \frac{q(\varphi)}{\varphi}$$

Hopenhayn: Definition of Stationary Equilibrium

- A stationary equilibrium is P, w, E, value function $V(\varphi)$, pricing/employment rules, cutoff φ^*, entrant mass M, and stationary distribution μ such that:

  1. Firms optimize statically, implying CES markup pricing
  2. $V(\varphi)$ and φ^* solve the incumbent's Bellman equation
  3. Free entry holds: $wc_e = \beta \int V(\varphi)\nu(d\varphi)$
  4. μ is stationary given entry, exit, and the Markov transition
  5. The CES price index is consistent with μ
  6. Labour and expenditure markets clear

- **Key point:** Hopenhayn's dynamic selection mechanism survives intact; CES changes the mapping from productivity to static profits

Hopenhayn: Existence and Uniqueness

- In the paper, existence is fairly easy: productivity space is compact, productivity process P is continuous, demand is sufficiently "nice" (satisfied by CES)
- Under CES, the stationary equilibrium is unique. In general, you need separability:

$$\pi(\varphi) = h(\varphi)g(E, P, w)$$

- Note that the stationary equilibrium might not feature entry and exit, but if entry costs are not too high then there will be positive entry and exit

Hopenhayn: What the Model Delivers

- A fully endogenous, stationary firm size distribution — generated purely from a Markov productivity process plus entry/exit, not assumed
- Endogenous entry and exit rates, consistent with observed "churn" even in steady state
- A framework for comparative statics:
  - Higher entry cost $c_e \Rightarrow$ fewer surviving firms, lower turnover
  - Policies that tax large/productive firms distort the exit cutoff and can reduce aggregate output

Hopenhayn: Predictions about Age

- Model has been written in terms of productivity φ. But under the right assumptions, can get nice predictions about firms by age. In particular:
  - Conditional on survival, productivity and size rise with age
  - Exit rates decline with age
  - Conditional on survival, growth declines with age
- Important: these findings come about purely from selection. There is no learning // innovation here!

Hopenhayn: Computation

- Solving the model numerically proceeds in several steps:
  1. Guess the wage w and normalize $P = 1$. Note $E = w \times \bar{L}$
  2. Solve firms' static problem to recover $p(\varphi), q(\varphi), \pi(\varphi)$
  3. Solve the firm's Bellman to recover $V(\varphi)$ and exit cutoff $\varphi^\star$
  4. Compute stationary distribution using productivity transition and exit rule
  5. Check free entry: adjust w until $\beta \mathbb{E}_\nu[V(\varphi; w)] = w c_e$
  6. Check labour market clearing: use aggregate labour demand to determine mass of firms